

Predicting Apprenticeship Outcomes

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Research Question

Background

The Department of Labor’s (DOL) Office of Apprenticeship oversees registered apprenticeship programs across American. These programs enable workers to earn income while developing critical job skills in both traditional and non-traditional apprenticeship occupations. At the end of their apprenticeship, workers earn a portable certification in their occupation while also experiencing structured wage growth as part of their path to becoming a journeyworker. Our analysis focuses on understanding and predicting the factors that result in these outcomes.

Research Questions

Our analysis focuses on two critical dimensions of apprenticeship outcomes: apprenticeship completion and apprentice exit wage. Developing models that determine which factors influence these outcomes will better enable investment and program structure that results in better outcomes for apprentices.

- Question 1: Which indicators (wage, occupational group, age, education level, race, and training model) are most useful in predicting whether an apprentice will complete their apprenticeship?
- Question 2: What do we predict an apprentice’s exit wage will be based on the apprentice’s occupational group, initial wage, age, gender, education level, race and training model?

Literature Review

To guide our research, we conducted a literature review. This literature review enabled us to develop hypotheses that better guided our research. The literature review emphasized the importance of a wide variety of predictors in apprentice outcomes, such as demographic factors, program design, and occupational factors. The literature review emphasized a considerable amount of nuance and interaction between factors, suggesting that a linear approach may not be feasible.

Data Resources

Our GitHub is at [this location](#). We used the U.S. Department of Labor’s public apprenticeship dataset, which is drawn from the Registered Apprenticeship Partner Information Database System (RAPIDS). This dataset contains information about all apprentices and their outcomes. It contains 77 variables. Since we are focused on outcome metrics, we used a time-lag subset of the data focused on apprentices who started their apprenticeships in 2018 and 2019. Given that apprenticeships can take up to 6 years to complete, this provides an appropriate time period for our research question. This dataset contained 159,052 observations.

Data Transformation & Methods

- We discovered significant data issues during exploratory data analysis, including extensive null values, category imbalances, and impossible/improbable values (e.g. wages over \$2,000 / hr).
- We cleaned the dataset by removing true missing values where no inference was possible, standardizing categorical variables, and resolving inconsistencies across occupational, demographic, and wage fields. We also addressed substantial NULL values in the training model variable and ensured all features were properly formatted for both parametric and tree-based modeling.
- We also created new variables where needed, such as expected term length in weeks (projected end date - start date) or completion status (this was based on the business definition of apprentice completion provided by the Office of Apprenticeship).
- Since we are focused on an outcome metrics, we cannot use any predictors associated with apprenticeship outcomes as predictors, most notably exit wage and completion status. Using these as predictors would undermine the model by introducing post-treatment bias.

Apprentice Completion (Classification)

Bottom Line

- The strongest predictors of apprentice completion are **higher starting wage** and **longer apprenticeship term length**. When significant non-linearity was introduced, **occupational group** became highly predictive.
- Beyond these, a wide variety of predictors (education level, race, ethnicity, sex) were demonstrated to be both statistically significant and substantial in their improvement to model results.
- Overall, while the Logistic Regression model provides greater explainability, it oversimplifies the phenomenon of apprentice completion. **Completion should be understood as a complex, nonlinear phenomenon where the outcome varies depending on interactions.** Model flexibility is key.

Approach

- The purpose of this classification modeling is to determine the factors associated with apprenticeship completion. This is an inference question with a mix of numeric and factor variables.
- Selected Methods: Decision Tree, Random Forest, Logistic Regression.** Rejected Methods: KNN, LDA, QDA, SVM: These methods have difficulty handling factor variables. SVM, Ridge, and Lasso: These methods are generally inferior to logistic regression when the goal is inference.
- We created a function to assess model performance across the key metrics (e.g. accuracy, F1)

Decision Tree

- Despite a large dataset (49,809), k-fold 10 cross validation determined that a simple decision tree was the best model. This tree had 3 terminal nodes.
- Results
 - Two variables were selected: Expected Weeks and Starting Wage.
 - Surprisingly, the model **identified longer apprenticeships as having higher completion rates**. Apprentices whose expected weeks were greater than 209.357 were classified into one decision space, whereby 67.74% were expected to complete.
 - Apprentices with lower a term lower 209.257 weeks were divided according to wage. Apprentices whose starting wage was more than \$21.54/hr completed 54.26% of the time, while apprentices who made less than that completed 32.92% of the time.
 - The accuracy of this model was **0.649** while the F1 Score was **0.558**.

Logistic Regression

- The Model consistently demonstrated issues with stability, resisting efforts to reduce bias. We investigated adding flexibility to the model by experimenting with the usage of splines on quantitative variables. However, we discovered that these added considerable variance to the model without significantly increasing performance.
- Ultimately, a simplified model was chosen. Almost all final predictors were statistically significant, with p values significantly lower than 0.05. However, this is a common effect of large datasets.
- Results
 - Ultimately, starting wage was the strongest continuous predictor of completion. Expected term length was the second strongest continuous predictor.
 - The accuracy of this model was **0.628**, while the F1 Score was **0.497**. It performed worse than the decision tree.

Random Forest

- The Logistic Regression model provided better explanation of predictors, but its performance was not good. We decided that this is due to significant non-linearities in the data. We decided to fit a Random Forest model to determine the gap between logistic regression and a more flexible model.
- Model performance was calibrated using functions to test the Out of Bag (OOB) error for different node sizes and mTry values. The final mTry was 8, while the final node size was 1. **This indicates that the tree benefited from high complexity and flexibility.** This echoes earlier findings about variable interactions
- Results
 - Starting wage was used in 22% of trees**, making it the most important predictor. Surprisingly, **the second most important predictor, Major Group Name (18% of trees) was not strongly represented in the logistic model.** This may indicate that it provides significant non-linear improvements for prediction. The remaining most significant predictors were term length (15%) and Educational attainment (11%).
 - The accuracy of this model was **0.717**, while the F1 Score was **0.659**. It performed significantly better than both the Decision Tree and the Logistic Regression model.

Charts and Figures

Apprentice Completion Charts

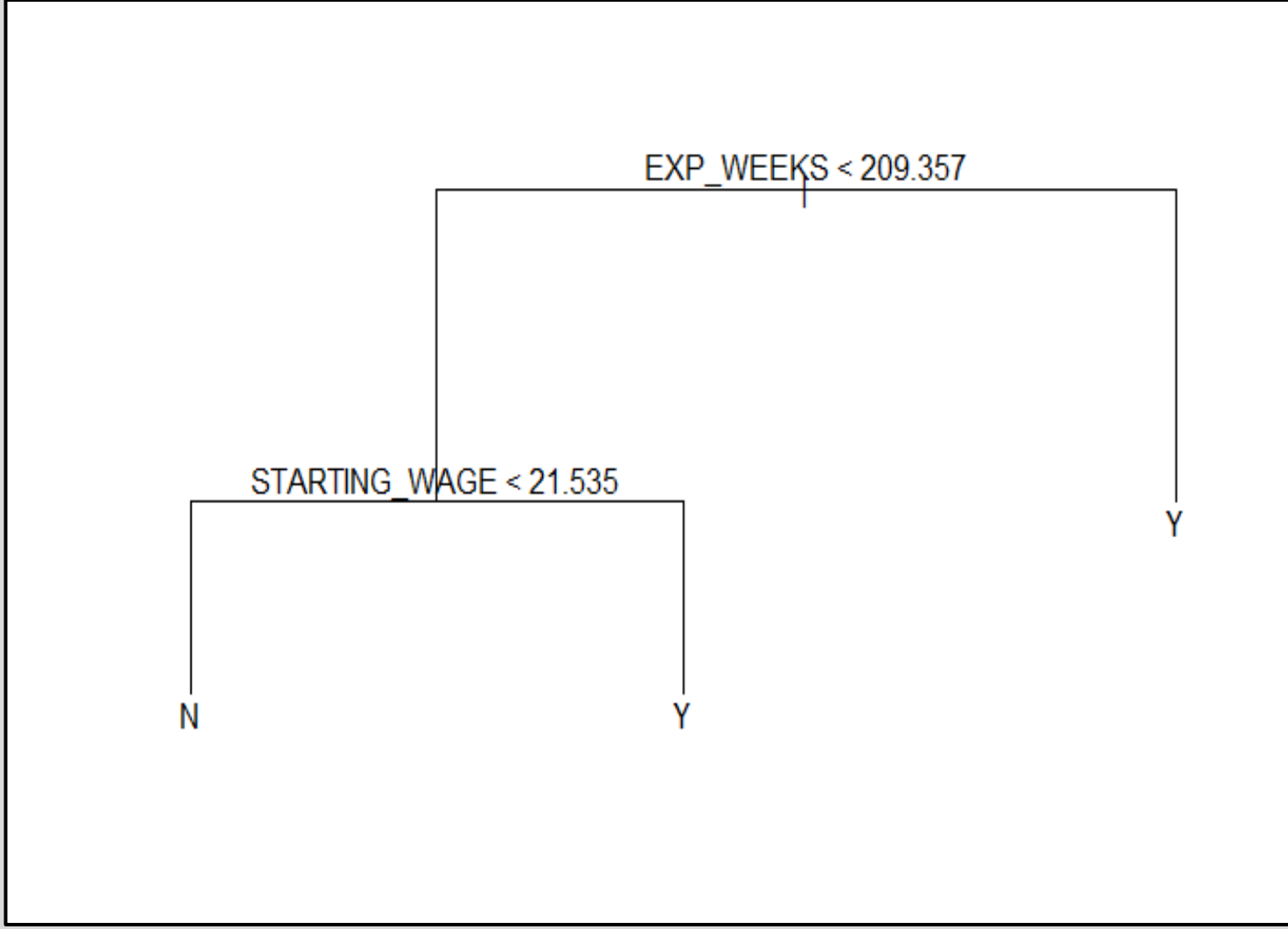


Fig 1: Basic Tree Model

Predictor	First Usage	Second Usage
EXP_WEEKS	0.32	29.58
STARTING_WAGE	0.74	14.13
MAJOR_GROUP_NAME	1.84	2.05
RACE_CD	3.05	3.55
OCCUP_TYPE	3.54	3.67
AGE_AT_START	3.61	3.62
ETHNICITY_CD_Rec	3.62	3.63
EDUCATION_CD_Rec	4.05	4.12
SEX_CD_Rec	4.31	4.32

Fig 2: RF Predictor Usage

Exit Wage Charts

Term	Estimate	Term	Estimate
(Intercept)	3.097394 ***	RACE_CDPacific Islander	0.168891
STARTING_WAGE	1.2190961 ***	RACE_CDBlack	-2.46109 ***
AGE_AT_START	-0.0891053 ***	RACE_CDAsian	1.609167 ***
SEX_CDFemale	-0.9128448 ***	RACE_CDNative American	1.324289 **
SEX_CDDid Not Self-Identify	-1.0620953 ***	RACE_CDNot Provided	-0.08754
ETHNICITY_CDNon-Hispanic or Latino	-0.6264771 ***	RACE_CDMulti-Race	0.156582
ETHNICITY_CDDid Not Self-Identify	-1.2725022 ***	OCCUP_TYPEHybrid	1.003357 **
EDUCATION_CDGED	2.62136 ***	OCCUP_TYPETime-Based	1.08653 ***
EDUCATION_CDHigh School graduate (including equivalency)	1.7039747 ***	MAJOR_GROUP_NAMEArts, Hospitality and Sales	-0.73754
EDUCATION_CDParticipant Did Not Self-Identify	0.3135474	MAJOR_GROUP_NAMEBusiness, Management and Office Support	-0.87469
EDUCATION_CDSome College or Associate's degree	1.7394015 ***	MAJOR_GROUP_NAMEHealthcare and Social Services	-1.43436 **
EDUCATION_CDBachelor's degree	1.6202737 ***	MAJOR_GROUP_NAMETrades and Manual Labor	1.960011 ***
EDUCATION_CDMasters degree	2.5650481 ***	MAJOR_GROUP_NAMEEducation and Public Services	-2.97423 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fig 3: Logistic Regression Coefficients

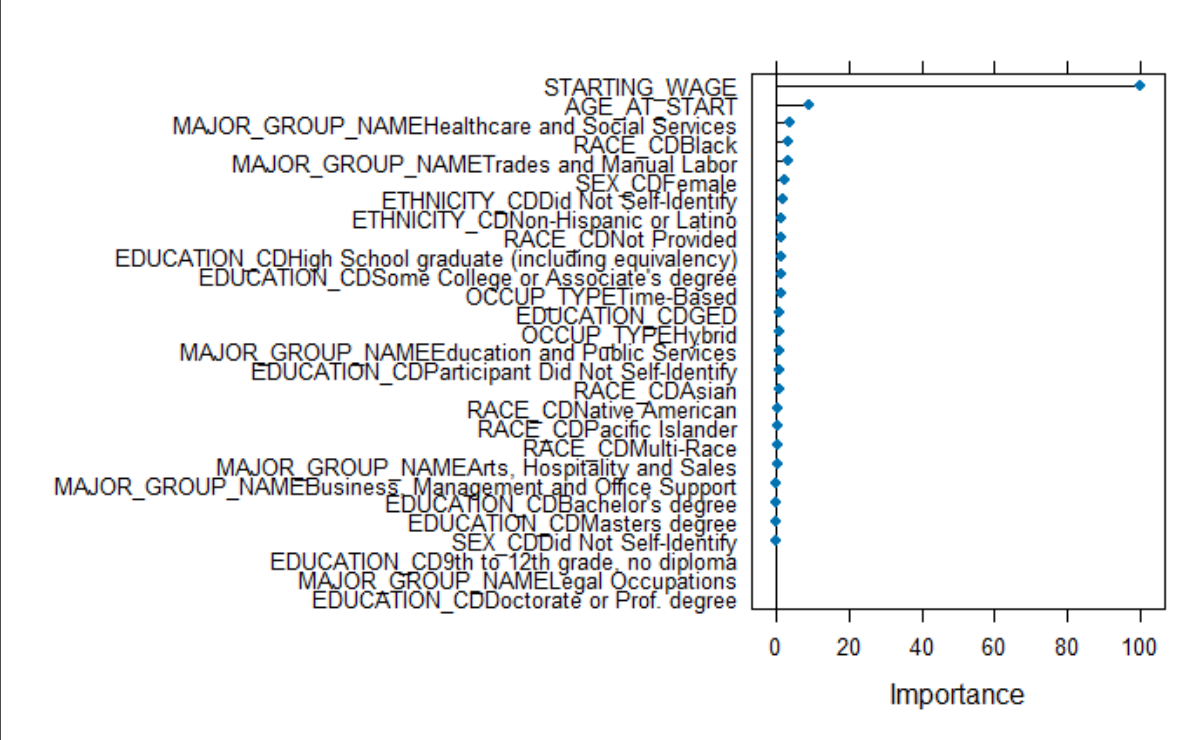


Fig 4: Random Forest Variable Importance

Model	Test MSE	Notes
Random Forest (Tuned)	70.59	Best model
Random Forest (Default)	73.75	Strong performance
Multiple Linear Regression	86.08	Good baseline, but underfits
Best Subset / Stepwise	86.10-86.15	No improvement over MLR
LASSO - λ -min	86.11	Similar to OLS (23 predictors)
Ridge	86.61	Slightly worse
LASSO - λ -1se	87.08	Similar to OLS (8 predictors)
PCR - 14 PC	87.33	Poor fit; categorical limitations
PLS - 8 PC	87.33	Similar to PCR
Regression Tree	90.81	Worst model (1 predictor)

Fig 5: Model Comparison

Exit Wage (Regression)

Data Source and Scope

- After filtering and cleaning for relevant features, the final analytic dataset contains 44,180 observations, with 8 predictors and the outcome (**EXIT_WAGE**).
- Continuous: **STARTING_WAGE** and **AGE_AT_START**
- Categorical: **MAJOR_GROUP_NAME**, **OCCUP_TYPE** (Training model), **EDUCATION_CD**, **SEX_CD**, **ETHNICITY_CD** and **RACE_CD**
- A 70%/30% validation split was used: Training set: 30,925 rows;Test set: 13,255 rows.This split is applied consistently across all models.
- Primary Evaluation Metric:** Test MSE/RMSE on held-out data

Model Selection Approaches:

- Adjusted R^2 , BIC, Mallows’ Cp, 5-fold CV using cv.glm
- All classical linear-model selection procedures produced very similar test MSE (approx. 85–86).
- No substantial improvement over the full MLR.

Ridge Regression

- Best λ :** 0.79. **CV MSE:** 84.91. **test MSE:** 86.61
- Ridge did not outperform OLS, indicating limited benefit from coefficient shrinkage.

LASSO Regression

- $\lambda_{\min}$:** test MSE = 86.11 and **λ_{1se} :** test MSE = 87.08
- LASSO selected 23 predictors at $\lambda_{\min}$ and performance was comparable to OLS.

Dimension Reduction Regression Models

- PCR:** Optimal components: 14 and **test MSE:** 87.33
- PLS:** Optimal components: **8** and **test MSE:** 87.33

Both performed worse than OLS because categorical predictors were removed.

Regression Tree

- Tree size:** 5 terminal nodes with only one predictor used: **Starting Wage**.
- Test MSE: 90.81** (worst of all models)

The tree model indicates strong dominance of STARTING_WAGE but fails to capture important interactions or nonlinearities.

Random Forest

- Random Forest default test: **test MSE:** 73.75, with percentage of variance in **Exit Wage** explained: **52.1%**.
- Tuned Random Forest (by ranger)
 - Best parameters (mtry = 8/min.node.size = 5/splitrule = variance/trees = 500)
 - Test MSE:** 70.59 (**Best overall model**)
 - Test RMSE** ~ 8.40
- Variable Importance (Scaled to 100):
 - Starting Wage, with importance = 100
 - Age at Start, with importance = 9
 - Major Group - Healthcare, Race: Black, Major Group - Trades, Sex: Female, with importance ~ 2.15 – 3.60.
 - Ethnicity, Education, Occupational Type, importance ~ 1 – 2.

The Random Forest model captures nonlinearities and interactions the linear models miss, resulting in the highest predictive accuracy by a wide margin.

Which Model Best Answers the Research Question?

Because the study’s goal is **prediction rather than inference**, model interpretability is secondary to predictive accuracy. Thus, the **best predictive model is the tuned Random Forest**, achieving the **Lowest Test MSE (70.59)**, with the **highest variance explained and the largest reduction in error compared with OLS** (~18% improvement in MSE).

Practical Implications

Starting Wage is the dominant predictor, suggesting that apprenticeship programs may reflect or reinforce preexisting labor market conditions. Occupational group matters substantially, and program designers should consider tailoring support and resources by sector. Demographic variables predict wage outcomes, which may signal a potential systemic inequities and opportunities for targeted support or policy intervention.

Conclusion

The recommendation is to use the **tuned Random Forest model** as the primary predictive engine for forecasting post-apprenticeship wages. Combine it with variable importance analysis to inform program design, highlight key predictors, and identify demographic or sector-specific areas in need of more exploration and attention.